Answer Key & Teacher Notes

Session 1 — From Table to Graph · Session 2 — Analyzing a Graph

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

From Table to Graph

TALK PROMPTS

- Why does the point (0, 0) belong on the dog-walking graph? What does it mean?
- Why must the input come first in an ordered pair?

PARTNER TASK

Babysitting Pay. One partner fills a table for $8 per hour (1, 2, 3 and 5 hours); the other writes the ordered pairs and plots them. Switch with Party Balloons at 16 balloons per bunch.

WATCH FOR

- Switching x and y in an ordered pair (plotting (75, 3) instead of (3, 75)).
- Using uneven spacing on an axis, so constant-rate points do not line up.
- Expecting every line to start at (0, 0), even when there is a starting fee.

SESSION 2 · TEACHER NOTES

Analyzing a Graph

TALK PROMPTS

- How does the steepness of a graph show the rate between two quantities?
- Why can't you read the y-value of any point as the rate?

PARTNER TASK

Two Pay Graphs. Job A's graph passes through (3, 24) and Job B's through (3, 15); both start at (0, 0). Each partner finds one rate, then together decide whose line is steeper and how far apart the pay is after 5 hours ($40 and $25, $15 apart).

WATCH FOR

- Reading the y-coordinate of a point as the rate (100 steps in 2 minutes read as 100 per minute).
- Dividing x by y instead of y by x.
- Comparing steepness on two graphs drawn with different scales.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.C.9	1	2	2	2	2	1	2	3	2	1	1
S1 · Form B	6.EE.C.9	1	2	2	2	2	1	2	3	2	1	1
S2 · Form A	6.EE.C.9	1	2	1	2	2	3	2	3	2	1	1
S2 · Form B	6.EE.C.9	1	2	1	2	2	3	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

ordered pair: Two numbers (*x*, *y*) that tell where a **point** is.

coordinates: The **two** numbers in an ordered pair.

axis: One of the two **number lines** on a graph.

plot: To **mark** a point on a graph.

- Each row of a table becomes an ordered pair: *x* **first**, then *y*.
- To plot (*x*, *y*), start at (0, 0). Move **right** *x*, then **up** *y*.

Example: The rows become (**1**, 4), (2, **8**) and (3, 12). Plot (1, 4): right 1, then up **4**. Answer: The points line up in a **straight** line.

YOU TRY

1. a. (1, 6) b. (2, 12) c. (4, 24)

2. The three points line up in a straight line through (0, 0).

EXIT TICKET

1. a. (2, 18) b. (5, 0)

2. (1, 2), (3, 6), (4, 8)

3. **B** (straight line)

PRACTICE FORM A

1. a. (3, 6) b. (5, 45) c. (0, 7) d. (8, 1)

2. 9, 18, 27, 45; points (1, 9), (2, 18), (3, 27), (5, 45) line up in a straight line through (0, 0)

3. (6, 300): after 6 seconds, the lift has risen 300 feet

4. She switched x and y. The nights (x) come first: (3, 75).

5. 6, 15, 24; points (2, 6), (5, 15), (8, 24) in a straight line

6. (5, 90): 5 rows have 90 seeds

7. a. y = 5x (multiply x by 5) b. a straight line through (0, 0) c. (7, 35)

8. (1, 6), (2, 8), (3, 10); the line meets the y-axis at (0, 4), the $4 fee

9. He switched x and y. Hours are x and dollars are y, so the points are (3, 15) and (5, 25).

10. **B** ((4, 20))

11. **C** ((3, 27))

Self-check word **GRAPH**

PRACTICE FORM B

1. a. (4, 28) b. (7, 2) c. (0, 5) d. (6, 54)

2. 8, 16, 24, 40; points (1, 8), (2, 16), (3, 24), (5, 40) line up in a straight line through (0, 0)

3. (8, 96): after 8 seconds, the elevator has risen 96 feet

4. He switched x and y. The days (x) come first: (3, 45).

5. 16, 48, 80; points (1, 16), (3, 48), (5, 80) in a straight line

6. (6, 84): 6 rows have 84 tiles

7. a. y = 4x (multiply x by 4) b. a straight line through (0, 0) c. (9, 36)

8. (1, 8), (2, 13), (3, 18); the line meets the y-axis at (0, 3), the $3 fee

9. She switched x and y. Cars are x and dollars are y, so the points are (2, 12) and (5, 30).

10. **C** ((6, 42))

11. **B** ((5, 40))

Self-check word **POINT**

GUIDED NOTES

rate of change: How much *y* changes when *x* goes up by **1**.

unit rate: The amount of *y* for **one** unit of *x*.

origin: The point **(0, 0)**, where the two axes cross.

steeper: A line that rises **faster** as you move right.

- Rate of change = change in *y* **÷** change in *x*.
- For a straight-line proportional relationship, use a point with x ≠ 0 and divide **y** by *x* for any point on it.

Example: Change in *y*: 15 − 6 = **9** dollars. Change in *x*: 5 − 2 = **3** hours. Divide: 9 ÷ 3 = **3**. Answer: The rate is $**3** per hour.

YOU TRY

1. a. 5 b. 7

2. (20 − 8) ÷ (5 − 2) = 4 miles per hour; 4 × 9 = 36 miles

EXIT TICKET

1. a. 8 b. 6

2. (48 − 24) ÷ (6 − 3) = $8 per bag; 8 bags: 8 × 8 = $64

3. **B** (9)

PRACTICE FORM A

1. a. 7 b. 5 c. 3 d. 9

2. a. 4 minutes cost $28 b. $7 per minute c. $21

3. a. 2 miles per hour b. 2 × 8 = 16 miles

4. a. 4 MB per song b. 5 songs use 20 megabytes

5. a. (28 − 14) ÷ (4 − 2) = 7 inches per week b. 7 × 9 = 63 inches

6. Job B is steeper: Job A pays $6 per hour and Job B pays $9 per hour, so Job B pays more each hour.

7. a. 25 gallons per hour b. (10, 250)

8. (3, 45) means 3 signs raise $45 in all, not per hour. $45 ÷ 3 = $15 per sign.

9. She used the y-value of (2, 100) as the rate but forgot to divide by 2 minutes. 100 ÷ 2 = 50, so the rate is 50 steps per minute.

10. **A** (8 per unit)

11. **B** (a greater rate)

Self-check word **RATES**

PRACTICE FORM B

1. a. 6 b. 4 c. 8 d. 11

2. a. 4 hours cost $24 b. $6 per hour c. $18

3. a. 3 miles per hour b. 3 × 7 = 21 miles

4. a. 5 MB per photo b. 6 photos use 30 megabytes

5. a. (12 − 6) ÷ (4 − 2) = 3 centimeters per week b. 3 × 11 = 33 centimeters

6. Job A is steeper: Job A pays $8 per hour and Job B pays $5 per hour, so Job A pays more each hour.

7. a. 20 liters per hour b. (12, 240)

8. (4, 36) means 4 boxes raise $36 in all, not per box. $36 ÷ 4 = $9 per box.

9. He used the y-value of (3, 120) as the rate but forgot to divide by 3 minutes. 120 ÷ 3 = 40, so the rate is 40 meters per minute.

10. **C** (7 per unit)

11. **B** (P has a greater rate)

Self-check word **STEEP**

SHEET 1 · FROM TABLE TO GRAPH

1. a. (4, 12) b. (0, 0) c. (6, 30) d. (9, 2)
2. $4, $8, $12, $20; points (1, 4), (2, 8), (3, 12), (5, 20) on a straight line through the origin
3. a. (5, 10) b. After 3 hours, Kofi has hiked 6 miles. c. 2 miles per hour
4. (1, 6), (3, 18), (4, 24). (0, 0) means 0 hours of renting costs $0.
5. He switched x and y. The x value (hours) comes first: plot (2, 10) and (4, 20).

SHEET 2 · ANALYZING A GRAPH

1. a. After 4 minutes, the pool has 60 gallons. b. 15 gallons per minute c. 135 gallons
2. a. 7 b. 8 c. 12
3. Unit rate: 18 ÷ 3 = 6 pages per hour; 7 hours: 6 × 7 = 42 pages
4. Job A: 24 ÷ 4 = $6 per hour. Job B: 18 ÷ 6 = $3 per hour. Line A is steeper, so Job A pays more each hour.
5. No. A proportional graph must pass through (0, 0); this one starts at (0, 10). (0, 10) means she had $10 saved at the start.
6. **C** (7)
7. **B** (Q is steeper; greater rate.)

October 2026 conceptual additions — teacher key

Session 1 · From Table to Graph

Points (0,0),(1,4),(3,12),(5,20). (3,12) means three notebooks cost $12. (2.5,10) fits the numerical rule but not whole-notebook purchasing.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Points (0,0),(1,4),(3,12),(5,20). (3,12) means three notebooks cost $12. (2.5,10) fits the numerical rule but not whole-notebook purchasing.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Analyzing a Graph

Points (0,5),(2,11),(4,17). Ratios 11/2=5.5 and 17/4=4.25 are not the $3 hourly rate because they include the fee. Output change 6 over 2 hours gives $3/hour.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Points (0,5),(2,11),(4,17). Ratios 11/2=5.5 and 17/4=4.25 are not the $3 hourly rate because they include the fee. Output change 6 over 2 hours gives $3/hour.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.